

Selenium E-Commerce Automation Testing Report

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Introduction

This project focuses on automating functional testing for an e-commerce web application using Selenium and Java. The purpose of the project was to automate repetitive test scenarios and validate core functionalities of the application.

Automation testing was performed on the [SauceDemo Website](#) to verify login functionality, cart operations, and checkout processes.

Objectives

- Automate functional test scenarios using Selenium WebDriver
- Validate login functionality using valid and invalid credentials
- Automate add-to-cart functionality
- Verify checkout workflow
- Gain hands-on experience in browser automation testing

Scope of Testing

The scope of this project includes:

- Login functionality testing
- Invalid login validation
- Add to cart functionality
- Checkout process automation

Testing was limited to functional automation testing only.

Tools and Technologies Used

Tool / Technology	Purpose
Selenium	Browser automation
Apache NetBeans	Development environment
ChromeDriver	Browser driver

Test Approach

- Selenium WebDriver was used to automate browser actions.
- Automated scripts were developed in Java.
- Functional scenarios were executed on the SauceDemo website.
- Positive and negative test cases were validated.
- Browser actions such as clicking, typing, navigation, and form submission were automated.

Automated Test Cases

Test Case ID	Test Scenario	Description	Expected Result	Status
TC01	Valid Login	Login using valid credentials	User logs in successfully	Pass
TC02	Invalid Login	Login using invalid credentials	Error message displayed	Pass
TC03	Add to Cart	Add product to shopping cart	Product added successfully	Pass
TC04	Checkout Process	Complete checkout workflow	Order completed successfully	Pass

Test Execution Details

- **TC01 – Valid Login Test**

Description: Automates login using valid credentials

Test Steps:

1. Open SauceDemo website
2. Enter valid username and password
3. Click Login button

Expected Result: User should log in successfully

Actual Result: User logged in successfully and navigated to products page

Status: Pass

- **TC02 – Invalid Login Test**

Description: Automates login using invalid credentials

Test Steps:

1. Open SauceDemo website
2. Enter invalid username and password
3. Click Login button

Expected Result: Error message should be displayed

Actual Result: Error message displayed successfully

Status: Pass

- **TC03 – Add to Cart Test**

Description: Automates adding a product to the shopping cart

Test Steps:

1. Login to application
2. Select product
3. Click Add to Cart
4. Open cart

Expected Result: Selected product should appear in cart

Actual Result: Product added successfully and displayed in cart

Status: Pass

- **TC04 – Checkout Test**

Description: Automates checkout workflow

Test Steps:

1. Login to application
2. Add product to cart
3. Open cart
4. Click Checkout
5. Enter customer information
6. Complete order

Expected Result: Checkout process should complete successfully

Actual Result: Order completed successfully

Status: Pass

Challenges Faced

- Understanding Selenium WebDriver setup and configuration
- Managing Selenium JAR dependencies
- Learning different locator strategies such as ID, Class Name, and CSS Selector
- Handling browser automation workflows
- Understanding automation testing concepts

Conclusion

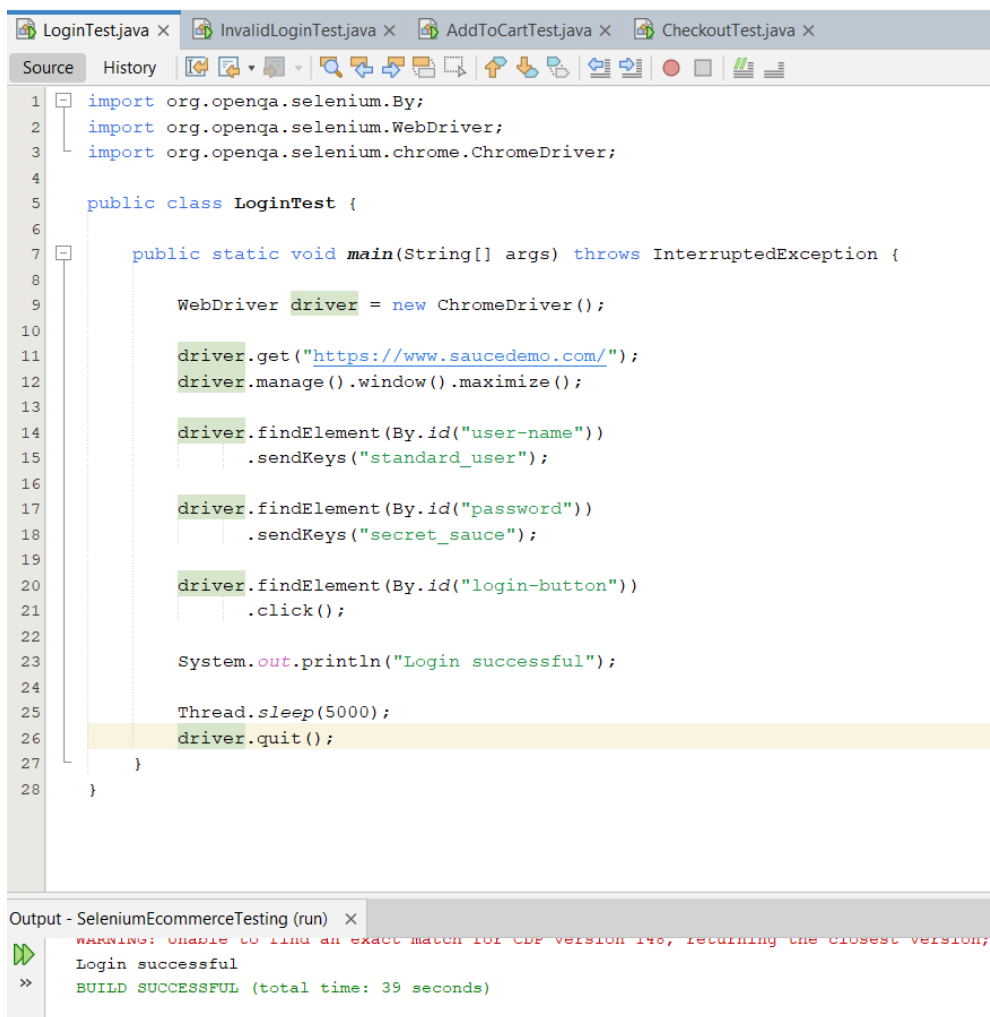
The project successfully automated core functionalities of an e-commerce website using Selenium WebDriver and Java. Automated testing improved efficiency by reducing manual execution of repetitive test cases.

The project provided practical experience in browser automation, element identification, functional validation, and automation workflow development.

Future Improvements

- Implement TestNG framework for better test management
- Add assertions for result validation
- Integrate automation testing with CI/CD pipelines

Screenshots



The screenshot displays an IDE with four tabs: LoginTest.java, InvalidLoginTest.java, AddToCartTest.java, and CheckoutTest.java. The 'Source' tab is active, showing the code for LoginTest.java. The code imports Selenium WebDriver and ChromeDriver, and implements a main method that performs a login on saucedemo.com. The output window at the bottom shows a warning about a missing CDF version, followed by 'Login successful' and 'BUILD SUCCESSFUL (total time: 39 seconds)'.

```
1 import org.openqa.selenium.By;
2 import org.openqa.selenium.WebDriver;
3 import org.openqa.selenium.chrome.ChromeDriver;
4
5 public class LoginTest {
6
7     public static void main(String[] args) throws InterruptedException {
8
9         WebDriver driver = new ChromeDriver();
10
11         driver.get("https://www.saucedemo.com/");
12         driver.manage().window().maximize();
13
14         driver.findElement(By.id("user-name"))
15             .sendKeys("standard_user");
16
17         driver.findElement(By.id("password"))
18             .sendKeys("secret_sauce");
19
20         driver.findElement(By.id("login-button"))
21             .click();
22
23         System.out.println("Login successful");
24
25         Thread.sleep(5000);
26         driver.quit();
27     }
28 }
```

Output - SeleniumEcommerceTesting (run) ×

```
WARNING: Unable to find an exact match for CDF version 146, returning the closest version;
Login successful
>> BUILD SUCCESSFUL (total time: 39 seconds)
```

Valid Login

Chrome is being controlled by automated test software

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Username

Password

Login

Accepted usernames are:

Password for all users:

standard_user
locked_out_user
problem_user
performance_glitch_user
error_user
visual_user

secret_sauce

Invalid Login

Chrome is being controlled by automated test software

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wrong_user

.....

Epic sadface! Username and password do not match any user in this service

Login

Accepted usernames are:

Password for all users:

standard_user
locked_out_user
problem_user
performance_glitch_user
error_user
visual_user

secret_sauce

Add to Cart

Chrome is being controlled by automated test software

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Your Cart

QTY	Description	
1	Sauce Labs Backpack carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.	\$29.99 Remove

Continue Shopping

Checkout



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Checkout Process

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saucedemo.com/checkout-complete.html

Chrome is being controlled by automated test software

Checkout: Complete!

Thank you for your order!

Your order has been dispatched, and will arrive just as fast as the pony can get there!

Back Home



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